AVTAAR

Client Overview

An interactive 3D avatar platform, built in-house, with a roadmap toward fully local conversation and deployment.

Version 1.0

Last updated: 12 October 2026

1 About Avtaar

Avtaar is an interactive 3D avatar platform built in-house. A rigged 3D character listens, speaks, looks at the user, and gestures during a conversation. Every part of the avatar's behavior is driven by our own engines, so we control its look, timing, and feel.

18	15	10
BEHAVIOR ENGINES	MOUTH SHAPES (VISEMES)	BODY POSES

What is built in-house

- The 3D avatar viewer and rendering pipeline (React Three Fiber).
- A set of 18 behavior engines that turn speech and conversation state into facial, eye, head, and body motion.
- Calibration panels for tuning eyes, head, gestures, and speech prosody on each avatar rig.
- An admin area for avatar appearance, lighting, poses, and behavior settings.

2 Engine List

The engines live in the avatar engine module. Each has one responsibility and does not write to the 3D model directly, except the output stage.

Group	Engine	Purpose
Core output	`AvatarEngine`	Owns model writes and timing for the avatar.
Core output	`AnimationMixer`	Blends animation channels by priority (idle, emotion, speech, blink, debug).
Core output	`FacialEngine`	Holds facial control weights and handles speech silence.
Speech	`LipSyncEngine`	Converts audio amplitude and visemes into mouth and jaw motion.
Speech	`ProsodyEngine`	Reads speech energy, pitch, and emphasis as semantic signals.
Conversation	`BehaviorController`	Switches the avatar between idle, listening, thinking, and speaking.
Emotion	`EmotionEngine`	Applies six emotions: neutral, happy, sad, angry, surprised, thinking.
Eyes	`EyeEngine`	Eye state for the eye rig.

Group	Engine	Purpose
Eyes	`EyeController`	Converts gaze targets into eye rotations.
Eyes	`SaccadeEngine`	Adds small, quick gaze movements.
Eyes	`BlinkCoordinator`	Owns blink timing and blink weight.
Gaze	`GazeBehaviorEngine`	Chooses the gaze target for each behavior state.
Head	`HeadMotionEngine`	Converts head intent into bounded head motion.
Head	`HeadEyeCoordinator`	Adds supporting head motion that follows the gaze.
Gestures	`GestureIntentEngine`	Chooses a gesture intent from speech and conversation signals.
Gestures	`GestureTimingEngine`	Decides when a gesture may start.
Gestures	`GestureEngine`	Plays small gesture poses into the animation channels.
Idle	`IdleEngine`	Natural idle body motion when no one is speaking.

Supporting modules

ConversationCoordinator handles playback events. **PoseController** drives 10 body poses with smooth transitions. **avatarState** saves each avatar's look.

3 Local LLM and Local Deployment

Our goal is to run the avatar's conversation entirely on local infrastructure.

Conversation model
A local LLM running on the client's own hardware, with no dependency on an external cloud service.

Hosting
The full platform deployed on the client's local server or workstation.

Voice
A local speech stack, to be selected together with the client.

> **Status:** This is the next major milestone and is currently in planning.

4 Lip-Sync: Where We Are

Lip-sync is not yet at the level of leading commercial avatar tools. The current system drives mouth shapes from audio energy and 15 visemes, and it is measured against timing reference clips.

Improvement phases

1. **Foundation**
   Audio-to-mouth timing within ±40 ms, and clear separation of A, E, O, M, F, and S mouth shapes.

2. **Advanced lip-sync**
   At least 95% accuracy on P, B, and M lip closures, median viseme timing error under 40 ms, and per-rig calibration profiles for jaw and lip gains.

3. **Ongoing**
   Progress updates are published as each phase is completed and measured.

> These are internal targets. Results will be reported only once they have been measured.

5 Update Cadence

This document will be updated as each engine, the local LLM, and each lip-sync phase is completed. Every revision will carry its date.

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